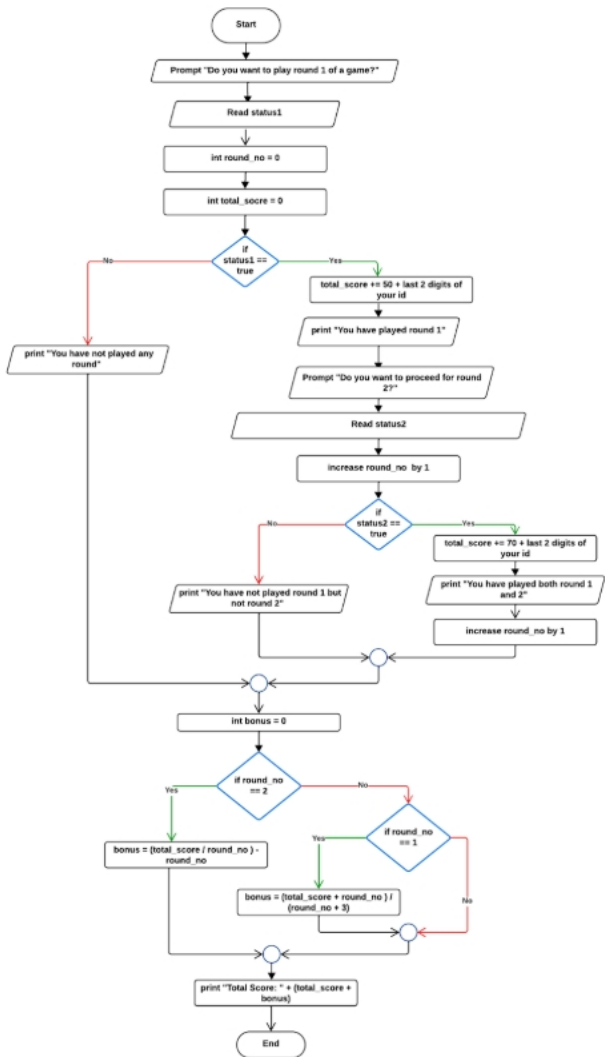




QUIZ II

BRAC UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CSE110: PROGRAMMING LANGUAGE I

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SECTION: 11

Question - 1: [20 points]

A debt collector is collecting debt with a timeframe of 3 years. The collector gives an opportunity for installment payment every month throughout these years. The borrower will start the payment monthly from the first year.

- **Payment each month = debt amount * payment rate /100**

If the borrower is able to pay off the debt at a certain month of a year, print that as output and stop further payment. If not, the borrower will have to continue paying.

After finishing the payment of a year, **condition for change in payment rate** the following year:

- If the borrower has paid off **more than 50% of the due payment** they had at the start of a particular year, the payment rate for the next year will be **incremented by 3**. Else, the rate will be **doubled for the following year**.

In case the debt cannot be repaid within 3 years, legal action will be taken!

You need to take the total debt amount and initial payment rate as **user inputs**.

Sample Input	Sample Output
Enter the debt amount: 50000.0 Enter the payment rate (%): 7	Amount paid in year 1: 42000.0. Total paid so far: 42000.0. Left to be paid: 8000. Debt has not been paid off after year : 1. Payment rate for next year (%) : 10 Debt has been paid off by month 2 of year 2
Enter the debt amount: 30000.0 Enter the payment rate (%): 1	Amount paid in year 1: 3600.0. Total paid so far: 3600.0. Left to be paid: 26400.0 Debt has not been paid off after year : 1. Payment rate for next year (%) : 2 Amount paid in year 2: 7200.0. Total paid so far: 10800.0 Left to be paid: 19200.0 Debt has not been paid off after year : 2. Payment rate for next year (%) : 4 Amount paid in year 3: 14400.0. Total paid so far: 25200.0 Left to be paid: 4800.0 Debt could not be covered in 3 years. Legal action will be taken now!

Explanation of 2nd example: At the start of year 1, the borrower started with 30000.0 debt and 1% payment rate. By that year, 3600.0 ($30000.0 * 1/100 * 12$) was covered, so 50% of 30000.0 could not be covered by the end of year 1. Thus, the rate for next year (year 2) was doubled to 2%. At the start of year 2, the borrower started with 26400.0 left to be paid. By the end of year 2, they could pay 7200.0 ($30000.0 * 2/100 * 12$), so 50% of 26400.0 could not be covered. The rate for next year (year 3) was doubled to 4%. 10800.0 is the total amount paid so far including year 1 and 2. ($3600.0 + 7200.0 = 10800.0$).

Solution:

```
import java.util.Scanner;
public class Midterm {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the debt amount: ");
        double debt = sc.nextDouble();
        System.out.println("Enter the installment rate: ");
        int rate = sc.nextInt();
        double totalPaid = 0;
        boolean flag = false;
        for (int i = 1; i <= 3; i++) {
            double startYearPaid = totalPaid;
            for (int j = 1; j <= 12; j++) {
                totalPaid += (debt) * (rate / 100.0);
                if (totalPaid >= debt) {
                    System.out.printf("Debt has been paid off by month %s of year %s.\n", j, i);
                    flag = true;
                    totalPaid = debt; //in case paid becomes greater than debt amount. Scenario not in
question.
                }
            }
        }
    }
}
```

question.

```
        break;
    }
}
if (flag) {
    break;
}
System.out.printf("Amount paid after year %s: %s. Left to be paid: %s\n", i, totalPaid,
(debt - totalPaid));
double paidThisYear = totalPaid - startYearPaid; //amount paid this year
if (i<=2) {
    if(paidThisYear >= ((debt - startYearPaid) * 50/100)) { //if amount paid this year >=
50% of remaining amount that was supposed to be paid this year
        rate += 3;
    }
    else {
        rate *= 2;
    }
    System.out.printf("Debt has not been paid off after year : %s.\nPayment rate for next
year : %s.\n", i, rate);
```

```
    }  
  }  
  if(!flag) {  
    System.out.println("Debt could not be covered in 3 years. Legal action will be taken  
now!");  
  }  
}  
}
```

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                    totalPaid = debt; //in case paid becomes greater than debt amount. Scenario not in
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                    break;
                }
            }
            if (flag) {
                break;
            }
            System.out.printf("Amount paid after year %s: %s. Left to be paid: %s\n", i, totalPaid,
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                    rate += 3;
                }
            }
        }
    }
}
```

50% of remaining amount that was supposed to be paid this year

```
        rate += 3;
    }
    else {
        rate *= 2;
    }
    System.out.printf("Debt has not been paid off after year : %s.\nPayment rate for next
year : %s.\n", i, rate);
```

```
    }
}
if(!flag) {
    System.out.println("Debt could not be covered in 3 years. Legal action will be taken
now!");
}
}
}
```

Quiz-3 Question

A BRAC University student is partaking in a match where he obtained some score. However, the commentary contains the student's mistaken score. Moreover, when stating the student's name more than once, we want to replace his name with the pronoun **'he'/'He'**.

Take the student's name (**String**), his correct score (**integer**) and the commentary (**String**) as user inputs and make the necessary corrections:

- **Replace wrong score** with the correct score in the commentary.
- **Replace repeated name at sentence start** with "He".
- **Replace repeated name elsewhere** with "he".

Note: You cannot use any built-in method except `length()`, `charAt()` and `equals()`. Your code must work for any given input — name and score.

Input: Enter the name: Ali

Enter the correct score: 70

Enter the commentary: Ali is a student at BRAC University. Ali played a match today where Ali scored 90. 90 is not a bad score!

Output: Ali is a student at BRAC University. He played a match today where he scored 70. 70 is not a bad score!

Input: Enter the name: Jaber

Enter the correct score: 80

Enter the commentary: Jaber played a match today where Jaber scored 190. 190 is not a bad score! Jaber is extraordinary!

Output: Jaber played a match today where he scored 80. 80 is not a bad score! He is extraordinary!

Solution---

```
import java.util.Scanner;

public class Quiz3 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the name: ");

        String name1 = sc.nextLine();

        System.out.print("Enter the original score: ");

        int score1 = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter the paragraph: ");

        String input = sc.nextLine();

        // "Jaber is a student at BRAC University. Jaber played a match today where Jaber
        scored 190. 190 is not a bad score!"

        String new_st = "";

        int count1 = 0;

        String temp = "";

        boolean flag = false;

        for (int i = 0; i < input.length(); i++) {

            if (input.charAt(i) == ' ') {

                if (temp.equals(name1)) {

                    if (count1 == 0) {

                        new_st += name1;

                    } else {

                        if (input.charAt(i - temp.length() - 1) == ' ' &&

                            (input.charAt(i - temp.length() - 2) == '.' ||

                                input.charAt(i - temp.length() - 2) == '?') ||
```

```

        input.charAt(i - temp.length() - 2) == '!')) {
            new_st += "He";
        } else {
            new_st += "he";
        }
    }
} else {
    new_st += temp;
}
new_st += " ";
temp = "";
count1++;
} else if (input.charAt(i) >= '0' && input.charAt(i) <= '9') {
    if (input.charAt(i - 1) >= '0' && input.charAt(i - 1) <= '9') {
        continue;
    }
    new_st += "" + score1;
} else {
    temp += input.charAt(i);
}
}
new_st += temp;
System.out.println(new_st);
}
}

```

Quiz-4 Question

You are given an array of integers which are passed to the `checkArrays()` method. Two conditions are to be satisfied for each element of the array:

- **Condition 1:** Each element of the array must exceed the average of even numbers on its left side.
- **Condition 2:** Each element of the array must exceed the average of odd numbers on its right side.

Note: You cannot use any built-in method. Your code must work for any given array.

Method Call:

```
int[] numbers = {10, 11, 4, 5, 13, 7, 40};  
checkArrays(numbers);
```

Output:

10 satisfies the given conditions.
11 satisfies the given conditions.
4 does not satisfy the given conditions.
5 does not satisfy the given conditions.
13 satisfies the given conditions.
7 does not satisfy the given conditions.
40 satisfies the given conditions.

Method Call:

```
int[] numbers = {2, 6, 50, 45, -3};  
checkArrays(numbers);
```

Output:

2 does not satisfy the given conditions.

6 does not satisfy the given conditions.

50 satisfies the given conditions.

45 satisfies the given conditions.

-3 does not satisfy the given conditions...

Solution----

```
public class Quiz4 {  
    public static void main(String[] args) {  
        int[] numbers1 = {2, 6, 50, 45, -3};  
        checkArrays(numbers1);  
    }  
  
    public static void checkArrays(int[] numbers1) {  
        for (int i = 0; i < numbers1.length; i++) {  
            double leftSum = 0, rightSum = 0;  
            int count1 = 0, count2 = 0;  
  
            for (int j = 0; j < i; j++) {  
                if (numbers1[j] % 2 == 0) {  
                    count1++;  
                    leftSum += numbers1[j];  
                }  
            }  
  
            for (int j = i + 1; j < numbers1.length; j++) {
```

```
        if (numbers1[j] % 2 != 0) {  
            count2++;  
            rightSum += numbers1[j];  
        }  
    }  
}
```

```
if (leftSum == 0) {  
    count1 = 1;  
}  
if (rightSum == 0) {  
    count2 = 1;  
}
```

```
double leftAvg = leftSum / count1;  
double rightAvg = rightSum / count2;
```

```
if (numbers1[i] > leftAvg && numbers1[i] > rightAvg)  
    System.out.println(numbers1[i] + " satisfies the given conditions.");  
else  
    System.out.println(numbers1[i] + " does not satisfy the given conditions.");  
}  
}  
}
```